

BE II Year Computer Engineering
Internal Test-3
2CERC3 Data Structure

Max Marks: 20
Time: 70 Mins

****Instructions****

Answer any 2 questions in each section. All questions carry equal marks

Section A: Attempt Any 2

1. What is a Tree Define it in context of Data Structure?
2. Find out the post-order traversal of given tree? Also write down the sequence of visited nodes.

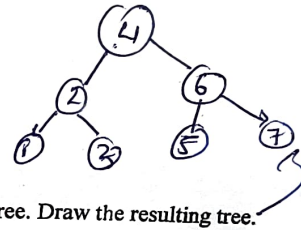
```
5
/\
3 8
/\
1 4
```

3. Write a short note on Tree Traversal. Give suitable examples.

Section B: Attempt Any 2

4. Explain the concept of AVL trees and why they are used?
5. Perform the necessary rotations to balance the following AVL tree:

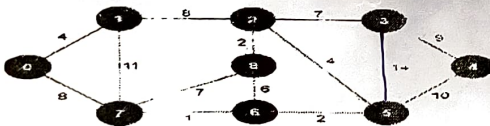
```
3
/\
2 7
/\
1 4
```



6. Insert the values 4, 2, 6, 1, 3, 5, and 7 into an initially empty binary search tree. Draw the resulting tree.

Section C: Attempt Any 2

7. Explain the Prim's algorithm for finding the minimum spanning tree with the help of example.
8. Differentiate between BFS and DFS with suitable example.
9. Apply Kruskal's algorithm to find the minimum spanning tree for the following graph:



Section D: Attempt Any 2

10. Apply Dijkstra's algorithm to find the shortest path from node 1 to every other node in the following graph:



11. Explain Adjacency list and matrix with example.
12. Compare Tree and Graph Data Structure and discuss their real life applicability.